

Statistical and SQL-Based Customer Analytics for E-Commerce Decision Making

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Abstract

This study presents a statistical and SQL-based analysis of customer behavior in an e-commerce dataset containing 60 customers and 26 original variables. The analysis integrates descriptive statistics, data-quality diagnostics, non-parametric inference, correlation analysis, customer-value scoring, churn-risk classification, segmentation, and business-oriented aggregation. The primary objective is to translate transactional and customer-level observations into statistically interpretable evidence for customer relationship management and marketing decisions.

The results indicate substantial heterogeneity in customer value and purchasing behavior. Total spending was strongly non-normal (Shapiro–Wilk $W = 0.708$, $p < 0.001$), while purchase count ($W = 0.957$, $p = 0.033$), average order value ($W = 0.915$, $p < 0.001$), age ($W = 0.938$, $p = 0.004$), and satisfaction score ($W = 0.879$, $p < 0.001$) also departed from normality. Spearman rank correlations showed a moderate positive association between purchase count and total spending ($\rho = 0.613$), and a weaker positive association between average order value and total spending ($\rho = 0.434$). Customer-value scoring divided the sample into three equally sized groups: 20 high-value, 20 medium-value, and 20 low-value customers. The high-value segment generated approximately 70.8% of total revenue.

A churn-risk rule based on more than 180 days since the last purchase classified 36 of 60 customers (60%) as at risk. Importantly, 12 customers were simultaneously classified as high-value and churn-risk, identifying a strategically critical retention segment. K-means clustering selected $k = 3$ as the preferred solution, with a silhouette score of 0.316; a Kruskal–Wallis test confirmed statistically significant differences between clusters ($H = 16.109$, $p = 0.000318$). The discount analysis did not provide evidence of a meaningful spending difference between discount and non-discount customers (Mann–Whitney $p = 0.4512$; Cohen’s $d \approx -0.022$; bootstrap 95% confidence interval for the mean difference: $[-2146.19, 2400.88]$).

Geographically, Mashhad produced the highest total revenue (64,118.44) and the highest mean customer spending (5,828.95), although its mean satisfaction score was only 2.91. Overall, the findings support targeted retention, value-based segmentation, geographically informed marketing, and selective rather than universal discounting.

Keywords: customer analytics; SQL; descriptive statistics; non-parametric statistics; customer lifetime value; churn risk; K-means; Mann–Whitney U; Kruskal–Wallis; e-commerce analytics.

1 Introduction

The rapid growth of e-commerce has increased the availability of customer-level transactional and behavioral data. However, the presence of data alone does not guarantee effective decision making. Managers need statistically defensible summaries that distinguish systematic patterns from random variation and identify customer groups with economically meaningful differences.

The present study uses SQL outputs as the analytical foundation and combines database aggregation with statistical reasoning. The analysis addresses five central business questions:

1. Which customers generate the greatest economic value?

2. Which customers are potentially vulnerable to churn?
3. Which geographical markets have the strongest revenue potential?
4. Is discount usage associated with higher customer spending?
5. Can customers be separated into statistically and behaviorally meaningful segments?

The analysis is deliberately designed around the distributional properties of the observed variables. Because several important measures violate normality, rank-based and distribution-free procedures are emphasized where appropriate.

2 Data and Analytical Framework

The analytical dataset contains 60 customers and 26 original variables. The principal variables include customer identifier, age, city, province, membership tier, purchase count, average order value (AOV), total spending, days since last purchase, discount usage, returned items, and satisfaction score.

Additional analytical variables were constructed from the raw attributes:

- customer value score;
- customer value segment;
- churn-risk indicator;
- churn-risk label;
- cluster membership.

The SQL layer was used for data validation, aggregation, ranking, cross-tabulation, and business summaries. Statistical analysis was then used to assess distributional characteristics, associations, group differences, and segmentation quality.

3 Data Quality Assessment

The first stage consisted of structural and logical quality checks. The dataset contained 60 customers and no duplicate records. The principal analytical variables did not contain material missingness that prevented the planned analyses.

A consistency check compared observed total spending with the expected amount implied by purchase count and average order value:

$$E_i = \text{PurchaseCount}_i \times \text{AOV}_i. \quad (1)$$

Two observations required additional review. Customer 1030 had observed total spending of 25,000.00 compared with an expected value of approximately 4,079.14, producing a discrepancy of about 20,920.86. Customer 1040 had observed spending of 126.00 compared with an expected value of approximately 1,280.44, producing a discrepancy of about 1,154.44.

These observations should be validated against the underlying transaction system before the data are used for financial reconciliation or automated decision rules.

Six records were also flagged for review in the returned-items analysis. Customers requiring review had substantially lower mean spending (596.10) and lower mean satisfaction (2.50) than customers without the flag (mean spending 4,040.29; mean satisfaction 3.04). This pattern suggests that return-related anomalies may also be associated with weaker customer experience.

4 Descriptive Statistical Analysis

The total observed revenue was 221,752.15, with a mean customer spending of approximately 3,695.87 and a mean AOV of approximately 213.16. The mean purchase count was approximately 17.38 transactions per customer.

The distribution of customer spending was highly heterogeneous. The largest spending values were substantially above the central tendency, which is consistent with the observed non-normal distribution and motivates the use of robust and non-parametric procedures.

4.1 Normality Assessment

Normality was evaluated using the Shapiro–Wilk test. The results are summarized in Table ??.

Table 1: Shapiro–Wilk normality tests

Variable	W	p -value	Interpretation
Age	0.938	0.004	Non-normal
Purchase Count	0.957	0.033	Non-normal
Average Order Value	0.915	< 0.001	Non-normal
Total Spending	0.708	< 0.001	Strongly non-normal
Last Purchase Days	0.970	0.142	No evidence against normality
Satisfaction Score	0.879	< 0.001	Non-normal

At the conventional $\alpha = 0.05$ significance level, normality was rejected for age, purchase count, AOV, total spending, and satisfaction. Consequently, rank-based statistics and robust summaries are more appropriate than relying exclusively on classical parametric assumptions.

5 Correlation Analysis

Because several variables were non-normal and the sample size was limited, Spearman’s rank correlation coefficient was used to quantify monotonic associations.

Table 2: Selected Spearman rank correlations

Variable Pair	Spearman ρ	Interpretation
Purchase Count – Total Spending	0.613	Moderate positive
AOV – Total Spending	0.434	Weak-to-moderate positive
Last Purchase Days – Satisfaction	-0.293	Weak negative
Age – Purchase Count	0.265	Weak positive

The strongest observed association was between purchase count and total spending ($\rho = 0.613$), indicating that customers who purchase more frequently tend to generate greater cumulative revenue. Nevertheless, the relationship is not deterministic, because spending also depends on order value.

The correlation between AOV and total spending was positive but weaker ($\rho = 0.434$), suggesting that both purchase frequency and order size contribute to customer value.

6 Customer Value Scoring

A composite customer-value score was constructed from standardized measures of total spending, purchase count, and AOV:

$$CVS = 0.50Z(\text{Total Spending}) + 0.30Z(\text{Purchase Count}) + 0.20Z(\text{AOV}). \quad (2)$$

The weighting scheme assigns the greatest contribution to cumulative economic value while retaining behavioral information from frequency and order size.

Customers were divided into three equal-sized quantile groups: High Value, Medium Value, and Low Value.

Table 3: Customer-value segments

Segment	n	Mean Spending	Total Revenue	Revenue Share
High Value	20	7,846.92	156,938.43	70.77%
Medium Value	20	2,278.88	45,577.59	20.55%
Low Value	20	961.81	19,236.13	8.68%

The concentration of approximately 70.77% of revenue within one-third of the customer base is a major managerial finding. It indicates a high degree of revenue concentration and supports differentiated customer relationship strategies.

7 Churn-Risk Analysis

Churn risk was operationalized using a recency threshold: customers with more than 180 days since their last purchase were classified as churn-risk customers.

Table 4: Churn-risk comparison

Group	n	Mean Spending	Mean Purchases	Mean Last-Purchase Days
Churn Risk	36	3,816.63	16.89	265.14
Active	24	3,514.73	18.13	102.08

The churn-risk group contains 36 of 60 customers, corresponding to a 60% observed risk rate under the selected rule. The most important strategic subgroup consists of the 12 customers who are simultaneously high-value and churn-risk.

This overlap is more informative than the overall churn rate because customer loss is economically asymmetric: losing a high-value customer is potentially more consequential than losing a low-value customer.

It should be emphasized that the 180-day rule is a classification criterion rather than a probabilistic survival model. It therefore identifies operational risk rather than estimating a causal probability of future churn.

8 Geographical Analysis

City-level aggregation revealed substantial heterogeneity in revenue.

Table 5: Selected city-level performance indicators

City	Customers	Total Revenue	Mean Spending	Mean Satisfaction
Mashhad	11	64,118.44	5,828.95	2.91
Tabriz	12	36,192.49	3,016.04	3.00
Karaj	7	26,189.81	3,741.40	2.86
Ahvaz	8	23,384.78	2,923.10	3.50
Isfahan	5	21,975.32	4,395.06	4.40
Shiraz	6	20,715.92	3,452.65	2.00
Rasht	5	15,168.54	3,033.71	2.60
Tehran	6	14,006.85	2,334.48	2.67

Mashhad generated the highest total revenue and the highest mean customer spending. However, its satisfaction score of 2.91 is not particularly strong. Thus, a pure revenue-maximization strategy may overlook customer-experience risk. Marketing investment in Mashhad should therefore be accompanied by service-quality monitoring.

Isfahan provides a contrasting profile: its mean satisfaction score of 4.40 is the highest in the dataset, while its customer base is relatively small. This combination may justify controlled customer-acquisition experiments.

9 Discount Effect Analysis

The relationship between discount usage and customer spending was examined using the Mann–Whitney U test because spending is strongly non-normal.

The mean spending was 3,642.42 for discount users and 3,736.74 for non-users. The observed difference was not statistically significant:

$$p = 0.4512. \quad (3)$$

The standardized effect size was approximately

$$d \approx -0.022, \quad (4)$$

which is practically negligible. A bootstrap 95% confidence interval for the mean spending difference was approximately

$$[-2146.19, 2400.88]. \quad (5)$$

Because the confidence interval includes zero and the effect size is close to zero, the data do not provide evidence that discount usage is associated with a systematic increase in spending.

This result should not be interpreted as proof that discounts never work. Rather, within this observational sample, there is insufficient evidence of a general spending uplift attributable to discount usage. Targeted experimentation or randomized A/B testing would be preferable for causal evaluation.

10 Membership and Channel Analysis

Membership tiers exhibited heterogeneous economic profiles. Gold customers generated the highest total revenue, 75,153.20, while Silver customers had the highest mean spending per customer, approximately 4,179.93. VIP customers had a higher mean satisfaction score of 3.40, whereas Silver customers had the lowest satisfaction score at 2.50.

Payment-channel analysis showed that cash payments generated the largest total revenue (80,781.87), followed by online wallet payments (71,402.63) and card payments (69,567.65).

Device analysis showed that web users generated approximately 78,876.06 in revenue, followed by Android users at 74,945.08 and iPhone users at 67,931.01. These figures describe observed channel performance and should not be interpreted causally without controlling for customer composition and exposure.

11 Age and Satisfaction Profiles

The 55+ age group exhibited the highest mean spending, approximately 6,592.10, and an average purchase count of 21.35. Younger age groups generally exhibited lower spending levels.

The relationship between satisfaction and spending was not monotonic enough to support a simple linear interpretation. Customers with a satisfaction score of 4 had mean spending of approximately 5,448.63, whereas the score-5 group had mean spending of approximately 2,677.52.

This pattern demonstrates why customer satisfaction should not be treated as a direct proxy for economic value. Satisfaction and spending represent different dimensions of customer behavior and should be modeled jointly.

12 Cluster Analysis

K-means clustering was used to identify behavioral customer segments. The preferred number of clusters was $k = 3$, with a silhouette coefficient of 0.316. Although the silhouette value indicates moderate rather than strong separation, it provides useful evidence of meaningful heterogeneity in the customer base.

Table 6: Cluster profiles

Cluster	n	Mean Age	Purchases	AOV	Spending	Satisfaction
0	6	58.50	30.33	348.60	14,188.76	3.17
1	23	43.96	8.30	309.07	2,603.76	2.65
2	31	40.65	21.61	115.78	2,475.26	3.19

Cluster 0 is particularly important because only six customers generate an average spending of 14,188.76. The total revenue attributed to this cluster is approximately 85,132.55, representing 38.39% of cluster-level revenue.

To test whether the clusters differ statistically, the Kruskal–Wallis test was applied. The result was

$$H = 16.109, \quad p = 0.000318. \quad (6)$$

At $\alpha = 0.05$, the null hypothesis of identical distributions across clusters is rejected. Thus, the segmentation is supported by statistically significant between-cluster differences, although the magnitude and business meaning of each cluster should still be interpreted substantively.

13 Machine-Learning-Based Churn Diagnostics

As a supplementary predictive analysis, Logistic Regression and Random Forest models were evaluated using stratified five-fold cross-validation. The predictor set included age, purchase count, AOV, total spending, satisfaction score, and returned items. Last-purchase days was

excluded from the predictive feature set to reduce target leakage because the churn label itself is constructed from recency.

The mean ROC-AUC for Logistic Regression was 0.564 ($SD = 0.084$), while Random Forest achieved 0.478 ($SD = 0.096$). These values indicate weak discrimination and suggest that the available behavioral variables do not provide a strong predictive signal for the operational churn label.

Random Forest feature importance ranked total spending (0.235), AOV (0.214), satisfaction (0.167), age (0.156), purchase count (0.149), and returned items (0.078) as the principal predictors.

The relatively weak cross-validated AUC values imply that the current dataset is better suited to descriptive segmentation and risk prioritization than to high-confidence individual churn prediction.

14 Integrated Statistical Interpretation

The statistical evidence supports four broad conclusions.

First, customer value is highly concentrated. The high-value segment contains only one-third of customers but generates approximately 70.77% of revenue. This concentration increases the economic importance of retention among high-value customers.

Second, the customer base exhibits substantial behavioral heterogeneity. The non-normal distributions, moderate correlations, and significant cluster differences all indicate that a single average customer profile would be an inadequate representation of the population.

Third, churn risk is operationally substantial. Sixty percent of the sample meets the selected recency-based risk criterion, and 12 customers combine high economic value with elevated recency risk. These customers should be prioritized before broad, undifferentiated retention campaigns.

Fourth, the evidence does not support universal discounting as a reliable revenue-growth mechanism. The Mann–Whitney test, near-zero effect size, and bootstrap confidence interval are all consistent with the conclusion that discount usage did not produce a detectable general spending difference in this sample.

15 Managerial Implications

The statistical findings translate into several actionable strategies:

1. **Prioritize high-value churn-risk customers.** The 12 customers in the intersection of the high-value and churn-risk groups represent the most economically sensitive retention segment.
2. **Use value-based rather than uniform marketing.** Since revenue is strongly concentrated, customer-level resource allocation should reflect expected economic value.
3. **Treat Mashhad as a high-potential but experience-sensitive market.** Its revenue performance is strong, but satisfaction should be monitored before substantially increasing acquisition expenditure.
4. **Use discounts selectively.** The lack of a significant spending difference suggests that blanket discounting is not justified by the current evidence.
5. **Investigate data anomalies before automated decisions.** The discrepancies identified for customers 1030 and 1040 may materially affect customer-value calculations and should be reconciled.

6. **Use cluster-specific campaigns.** The statistically significant differences between clusters support differentiated communication, loyalty benefits, and retention strategies.
7. **Improve predictive data collection.** The weak cross-validated ROC-AUC values indicate that additional temporal, behavioral, and interaction-level features would be useful for future churn modeling.

16 Limitations

Several limitations should be considered when interpreting the results. First, the sample size is small ($n = 60$), which limits statistical power and the stability of predictive models. Second, the analysis is observational; therefore, associations between discount usage, spending, satisfaction, and churn risk should not be interpreted as causal effects.

Third, the churn definition is rule-based rather than derived from a time-to-event survival model. Fourth, the K-means solution provides moderate rather than strong separation, as reflected by a silhouette score of 0.316. Finally, data-quality discrepancies in selected records may influence individual-level rankings and should be resolved before operational deployment.

17 Overall Conclusion

This study demonstrates how SQL-based customer analytics can be extended into a statistically grounded decision-support framework. The analysis of 60 customers revealed pronounced heterogeneity in purchasing behavior, customer value, geographic performance, and churn risk.

The strongest finding is the concentration of economic value: the high-value third of the customer base accounts for approximately 70.77% of total revenue. At the same time, 36 customers, representing 60% of the sample, satisfy the operational churn-risk criterion. The intersection of these two dimensions identifies 12 high-value customers who are also at risk of churn. From a managerial perspective, this group represents the most immediate retention opportunity because the potential economic cost of losing these customers is substantially greater than that of losing low-value customers.

The inferential analysis further shows that customer behavior cannot be adequately summarized by simple means or normal-theory assumptions. Several variables are significantly non-normal, motivating the use of Spearman correlation, Mann–Whitney U, and Kruskal–Wallis procedures. The clustering analysis identifies three statistically distinguishable customer profiles, while the weak predictive AUC values indicate that the current feature set is insufficient for reliable individual-level churn prediction.

The discount analysis is particularly important for managerial decision-making. No statistically significant spending difference was detected between discount users and non-users ($p = 0.4512$), the estimated effect size was negligible, and the bootstrap confidence interval contained zero. Consequently, discounts should be viewed as a tactical instrument requiring targeting and experimentation rather than as a general-purpose mechanism for increasing customer spending.

Finally, Mashhad emerged as the strongest revenue market, but its moderate satisfaction level highlights the need to balance acquisition with customer-experience improvement. Overall, the evidence supports a data-driven strategy centered on value-based segmentation, targeted retention, selective incentives, geographic prioritization, and stronger data-quality controls. Future research should extend the analysis with transaction-level temporal data, survival analysis, causal experiments for discount effectiveness, and richer behavioral features to improve predictive customer analytics.

Key Statistical Indicators

Table 7: Summary of principal analytical indicators

Indicator	Value
Sample size	60 customers
Total revenue	221,752.15
Mean customer spending	3,695.87
Mean AOV	213.16
High-value customers	20
High-value revenue share	70.77%
Churn-risk customers	36 (60%)
High-value and churn-risk customers	12
Best city by total revenue	Mashhad
Mashhad total revenue	64,118.44
Mashhad mean spending	5,828.95
Discount test: Mann–Whitney p	0.4512
Discount effect size: Cohen’s d	≈ -0.022
Discount mean-difference bootstrap 95% CI	$[-2146.19, 2400.88]$
Optimal K-means k	3
Silhouette score	0.316
Kruskal–Wallis H	16.109
Kruskal–Wallis p	0.000318
Logistic Regression mean ROC-AUC	0.564
Random Forest mean ROC-AUC	0.478